

DATA STRUCTURES

CSA0301

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1. Student Marks Record:

```
#include <stdio.h>

int main() {
    int marks[5][4];
    int total[5];
    float average[5];
    int i, j;
    int topper = 0;

    // Input marks
    for(i = 0; i < 5; i++) {
        printf("Enter marks for Student %d:\n", i + 1);
        for(j = 0; j < 4; j++) {
            printf("Subject %d: ", j + 1);
            scanf("%d", &marks[i][j]);
        }
    }

    // Display marks
    printf("\nMarks of Students:\n");
    for(i = 0; i < 5; i++) {
        printf("Student %d: ", i + 1);
        for(j = 0; j < 4; j++) {
            printf("%d ", marks[i][j]);
        }
        printf("\n");
    }

    // Calculate total and average
    printf("\nTotal and Average Marks:\n");
    for(i = 0; i < 5; i++) {
        total[i] = 0;
        for(j = 0; j < 4; j++) {
            total[i] += marks[i][j];
        }
        average[i] = total[i] / 4.0;
        printf("Student %d - Total: %d Average: %.2f\n",
            i + 1, total[i], average[i]);
    }

    // Find topper
    for(i = 1; i < 5; i++) {
        if(total[i] > total[topper]) {
            topper = i;
        }
    }
}
```

```

    }
}

printf("\nTopper is Student %d with Total Marks = %d\n",
    topper + 1, total[topper]);

return 0;
}

```

2. Employee Salary Record:

```

#include <stdio.h>
#include <stdlib.h>

int main()
{
    int i, j;
    float **salary, total;

    /* Dynamic allocation for 6 employees */
    salary = (float **)malloc(6 * sizeof(float *));
    if (salary == NULL)
    {
        printf("Memory allocation failed.\n");
        return 1;
    }

    /* Allocate memory for 3 months for each employee */
    for (i = 0; i < 6; i++)
    {
        salary[i] = (float *)malloc(3 * sizeof(float));
        if (salary[i] == NULL)
        {
            printf("Memory allocation failed.\n");
            return 1;
        }
    }

    /* Input salaries */
    printf("Enter the monthly salaries of 6 employees for 3 months:\n\n");

    for (i = 0; i < 6; i++)
    {
        printf("Employee %d:\n", i + 1);
        for (j = 0; j < 3; j++)
        {
            printf("  Month %d: ", j + 1);
            scanf("%f", &salary[i][j]);
        }
        printf("\n");
    }

    /* Display salary matrix */
    printf("\nSalary Matrix:\n\n");
    printf("Employee\tMonth 1\t\tMonth 2\t\tMonth 3\n");
}

```

```

for (i = 0; i < 6; i++)
{
    printf("%d\t\t", i + 1);
    for (j = 0; j < 3; j++)
    {
        printf("%.2f\t\t", salary[i][j]);
    }
    printf("\n");
}

/* Calculate and display total salary for each employee */
printf("\nTotal Salary Paid to Each Employee:\n\n");

for (i = 0; i < 6; i++)
{
    total = 0;

    for (j = 0; j < 3; j++)
    {
        total += salary[i][j];
    }

    printf("Employee %d : %.2f\n", i + 1, total);
}

/* Free allocated memory */
for (i = 0; i < 6; i++)
{
    free(salary[i]);
}

free(salary);

return 0;
}

```

3. Hospital Patient Billing:

```

#include <stdio.h>
#include <stdlib.h>

int main()
{
    int i, j;
    float **bill, total;

    /* Dynamic memory allocation for 5 patients */
    bill = (float **)malloc(5 * sizeof(float *));
    if (bill == NULL)
    {
        printf("Memory allocation failed.\n");
        return 1;
    }
}

```

```

/* Allocate memory for 4 billing categories */
for (i = 0; i < 5; i++)
{
    bill[i] = (float *)malloc(4 * sizeof(float));
    if (bill[i] == NULL)
    {
        printf("Memory allocation failed.\n");
        return 1;
    }
}

/* Input treatment charges */
printf("Enter the treatment charges for 5 patients:\n\n");

for (i = 0; i < 5; i++)
{
    printf("Patient %d:\n", i + 1);

    printf("Consultation: ");
    scanf("%f", &bill[i][0]);

    printf("Lab Test: ");
    scanf("%f", &bill[i][1]);

    printf("Medicine: ");
    scanf("%f", &bill[i][2]);

    printf("Room Rent: ");
    scanf("%f", &bill[i][3]);

    printf("\n");
}

/* Display billing matrix */
printf("\nTreatment Charges Matrix:\n\n");
printf("Patient\tConsultation\tLab Test\tMedicine\tRoom Rent\n");

for (i = 0; i < 5; i++)
{
    printf("%d\t", i + 1);
    for (j = 0; j < 4; j++)
    {
        printf("%.2f\t", bill[i][j]);
    }
    printf("\n");
}

/* Calculate and display total bill */
printf("\nTotal Bill for Each Patient:\n\n");

for (i = 0; i < 5; i++)
{
    total = 0;

    for (j = 0; j < 4; j++)
    {

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        total += bill[i][j];
    }

    printf("Patient %d : %.2f\n", i + 1, total);
}

/* Free allocated memory */
for (i = 0; i < 5; i++)
{
    free(bill[i]);
}

free(bill);

return 0;
}

```

4. Cricket Scoreboard:

```

#include <stdio.h>
#include <stdlib.h>

int main()
{
    int i, j;
    int **runs;
    int total, highest = 0, highestPlayer = 0;

    /* Dynamic memory allocation for 6 players */
    runs = (int **)malloc(6 * sizeof(int *));
    if (runs == NULL)
    {
        printf("Memory allocation failed.\n");
        return 1;
    }

    /* Allocate memory for 5 matches */
    for (i = 0; i < 6; i++)
    {
        runs[i] = (int *)malloc(5 * sizeof(int));
        if (runs[i] == NULL)
        {
            printf("Memory allocation failed.\n");
            return 1;
        }
    }

    /* Input runs */
    printf("Enter the runs scored by 6 players in 5 matches:\n\n");

    for (i = 0; i < 6; i++)
    {
        printf("Player %d:\n", i + 1);

        for (j = 0; j < 5; j++)
        {

```

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        printf(" Match %d: ", j + 1);
        scanf("%d", &runs[i][j]);
    }

    printf("\n");
}

/* Display scorecard */
printf("\nScorecard:\n\n");
printf("Player\tMatch1\tMatch2\tMatch3\tMatch4\tMatch5\n");

for (i = 0; i < 6; i++)
{
    printf("%d\t", i + 1);

    for (j = 0; j < 5; j++)
    {
        printf("%d\t", runs[i][j]);
    }

    printf("\n");
}

/* Calculate total runs and highest scorer */
printf("\nTotal Runs of Each Player:\n\n");

for (i = 0; i < 6; i++)
{
    total = 0;

    for (j = 0; j < 5; j++)
    {
        total += runs[i][j];
    }

    printf("Player %d : %d\n", i + 1, total);

    if (total > highest)
    {
        highest = total;
        highestPlayer = i + 1;
    }
}

printf("\nHighest Scorer: Player %d\n", highestPlayer);
printf("Total Runs: %d\n", highest);

/* Free allocated memory */
for (i = 0; i < 6; i++)
{
    free(runs[i]);
}

free(runs);

return 0;
}

```

5. Library Book Issue Record:

```
#include <stdio.h>
#include <stdlib.h>

int main()
{
    int i, j;
    int **books;
    int total;

    /* Dynamic memory allocation for 5 departments */
    books = (int **)malloc(5 * sizeof(int *));
    if (books == NULL)
    {
        printf("Memory allocation failed.\n");
        return 1;
    }

    /* Allocate memory for 4 weeks */
    for (i = 0; i < 5; i++)
    {
        books[i] = (int *)malloc(4 * sizeof(int));
        if (books[i] == NULL)
        {
            printf("Memory allocation failed.\n");
            return 1;
        }
    }

    /* Input books issued */
    printf("Enter the number of books issued by 5 departments for 4 weeks:\n\n");

    for (i = 0; i < 5; i++)
    {
        printf("Department %d:\n", i + 1);

        for (j = 0; j < 4; j++)
        {
            printf("  Week %d: ", j + 1);
            scanf("%d", &books[i][j]);
        }

        printf("\n");
    }

    /* Display data */
    printf("\nBooks Issued Data:\n\n");
    printf("Department\tWeek1\tWeek2\tWeek3\tWeek4\n");

    for (i = 0; i < 5; i++)
    {
        printf("%d\t\t", i + 1);
```

```

        for (j = 0; j < 4; j++)
        {
            printf("%d\t", books[i][j]);
        }

        printf("\n");
    }

    /* Calculate and display total books issued */
    printf("\nTotal Books Issued by Each Department:\n\n");

    for (i = 0; i < 5; i++)
    {
        total = 0;

        for (j = 0; j < 4; j++)
        {
            total += books[i][j];
        }

        printf("Department %d : %d\n", i + 1, total);
    }

    /* Free allocated memory */
    for (i = 0; i < 5; i++)
    {
        free(books[i]);
    }

    free(books);

    return 0;
}

```